

Data Mining 2 Project

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1 DATA UNDERSTANDING AND PREPARATION

The goal of this project is to analyze two datasets providing different kinds of information about movies, TV shows, and other forms of visual entertainment.

In the following section, we'll focus on the tabular dataset, while the overview about the time series dataset composition and preparation can be found in Section 6.

This dataset is an extension of the one proposed in DM1: the records include key information about titles, insights into critical aspects such as awards and reviews, as well as ratings and further data about the title's production.

1.1 Data Semantics, Duplicates and Train/Test Split

The tabular dataset contains 149,531 records and 32 attributes; the nine attributes that were added to the original set from the DM1 dataset are reported in Tab. 1. This makes a total of nine categorical attributes, three binary attributes and 20 numerical attributes. Among the latter, only *runtimeMinutes* and *averageRating* can be classified as continuous, as the others mostly represent counts.

A first data quality check revealed only ten genuine duplicates, i.e. records sharing all feature values, which we dropped.

We then split the dataset, reserving 70% for training, and the remaining 30% for testing. As to avoid data leakage, we performed all subsequent data exploration exclusively on the training set. At the end of the data preparation stage, we applied all transformation and cleaning steps to the test set as well.

1.2 First Global Exploration of the Training Set

A first global analysis of the dataframe's summary and descriptive statistics revealed missing values for *endYear* (100,729), *runtimeMinutes* (28,185), *countryOfOrigin* (27,961) and *genres* (1,887). Since most records are missing an *endYear*, and the feature is strongly correlated to *startYear*, we dropped it.

Since a great majority of records had no information about sound mixes, we decided to drop this feature as well.

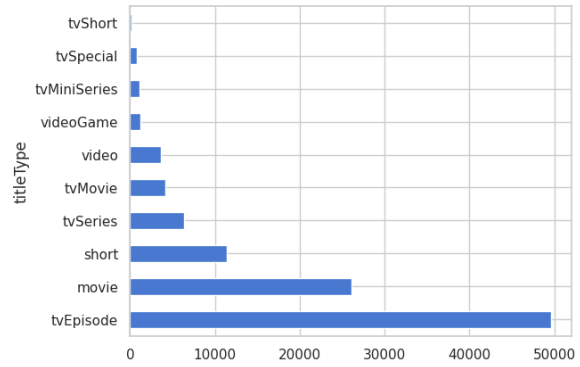


Figure 1: Distribution of *titleTypes*.

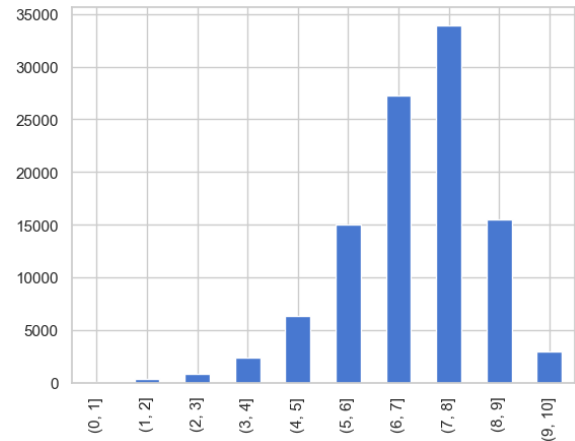


Figure 2: Barplot for *rating*.

bestRating, *worstRating*, and *isRatable* were once again confirmed to be uninformative, as they only take one value for all records (10, 1, and 1, respectively), and were therefore dropped. We also dropped *ratingCount*, as it was strongly correlated to *numVotes*, and *isAdult*, because it's redundant w.r.t. the *adult* category in *genres*.

Checking the values for the only remaining original binary attribute, *canHaveEpisodes*, we noticed that only two records with a *titleType* different from *tvSeries* and *tvMiniSeries* have a positive value. We corrected them: the first had been registered as a *tvMovie* instead of *tvSeries*, while the second was a short with no episodes.

Lastly, we removed the string '\\N', which was prepended to all lists of *regions*, making the feature consistent with *numRegions*.

1.3 Distribution of Categorical Variables and Statistics

In this subsection, we studied the univariate and bivariate distribution of categorical variables, which all turned out to be unbalanced.

Feature	Description	Type
castNumber	Total Number of Cast individuals present within the IMDb title page.	Discrete. Ratio-scaled.
companiesNumber	Total Number of companies that worked for the title.	Discrete. Ratio-scaled.
regions	The regions for this version of the title. Some titles can belong to more than one class.	Categorical. 171 unique classes.
averageRating	Weighted average of all the individual user ratings.	Continuous. Ratio-scaled.
externalLinks	Total Number of External Links the title has within the IMDb page.	Discrete. Ratio-scaled.
quotesTotal	Total Number of quotes the title has within the IMDb page.	Discrete. Ratio-scaled.
writerCredits	Total number of writer credits of the title.	Discrete. Ratio-scaled.
directorCredits	Total number of director credits of the title.	Discrete. Ratio-scaled.
soundMixes	Technical specification of the sound mixes available for the title. Some titles can belong to more than one class.	Categorical. 59 unique classes.

Table 1: New features in the extended dataset.

Differently from the DM1 dataset, the most popular *titleType* is *tvEpisode* (47% of the records), followed by *movie* (25%) and *short* (11%); *tvSpecial* (0.79%) and *tvShort* (0.17%) are the classes with the lowest support (cf Fig 1). With regard to *genres*, the two most popular classes are by far *Drama* and *Comedy*, although it depends on *titleType*. Binned *rating* follows a left-skewed distribution, with the mode being the [7,8) class (Fig. 2). The most popular *countryOfOrigin* and *region* is the US, but both features have a substantial amount of missing values: *nan* is the second most popular value for both.

We dropped *soundMixes*: not only are most of its 59 unique values very rare, it has too many missing values to provide useful information.

1.4 Distribution of Numeric Variables and Statistics

Before analyzing numeric variables, we added the same features we had added to the DM1 dataset:

- *numCountriesOfOrigin*: the number of countries listed for the record for the variable *countryOfOrigin*. For the training set, the values range from 1 to 10.
- *numGenres*: the number of genres listed in *genres*. For the training set, the values range from 1 to 3.
- *reviewsTotal*: the sum of *criticReviewsTotal* and *userReviewsTotal*.
- *criticReviewsRatio*: the proportion of review from critics (*criticsReviewTotal*) on the total number of reviews of a record (*reviewsTotal*). For unreviewed records, it is set to zero.
- *awardsAndNominations*: the sum of *awardWins* and *awardNominationsExcludeWins*.

A pairplot of all numeric variables, with KDE graphs on the diagonal, allowed us to gain a

global view of both the features' univariate distribution and their pairwise correlations. Given the size of the dataset, we resorted to using a random sample of 20,000 records. The great majority of the variables has a right-skewed distribution (*writerCredits*, *numRegions*, *awardWins*, *totalImages*, *criticReviewsTotal*, *runtimeMinutes*, *userReviewsTotal*, *totalCredits*, *totalVideos*, *castNumber*, *numVotes*, *quotesTotal*, *awardNominationsExcludeWins*, *companiesNumber*, *externalLinks*, *awardsAndNominations*, *numCountriesOfOrigin*, *reviewsTotal*). *averageRating*, of course, follows the same left-skewed distribution observed for its discretized version, *rating* (cf. Fig. 2).

We observed that some records have an implausibly long runtime, probably due to the inconsistent strategy adopted for the computation of the runtime for serialized content, for which sometimes the total runtime of a series is registered, rather than the average episode runtime. In order to avoid this problem, we removed the runtime of all *tvSeries*, *tvMiniSeries*, and *tvEpisodes* for which the value exceeded the 3rd quartile + 1.5 interquartile range for its specific category (105, 382.5 and 84 minutes, respectively); a total of 1560 records in the training set were affected.

A similar problem was found for *writerCredits*: records with suspiciously high values are likely TV episodes for which all writers of the entire TV series were erroneously credited. Therefore, we decided to remove the values for *writerCredits* from 2100 *tvEpisodes*, as they exceeded the 3rd quartile + 1.5 interquartile range (8.5 credits).

1.5 Variable Transformation, Pairwise Correlations and Elimination of Variables

We log-transformed all right-skewed variables, and used the Shapiro-Wilk test to determine whether they now resembled a normal distribution. None of them did, therefore we used Spear-

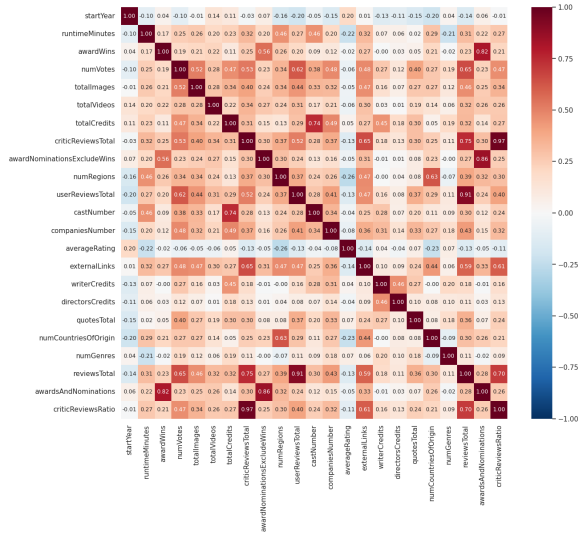


Figure 3: Correlation heatmap before attribute removal.

Original feature	Boolean version	% True
awardWins	hasAwards	8.16
awardsAndNominations	hasAwardsOrNominations	12.25
totalVideos	hasVideos	7.93
criticReviewsTotal	hasCriticsReviews	24.88
userReviewsTotal	hasUserReviews	36.98
directorsCredits	moreDirectorsCredits	9.89
quotesTotal	hasQuotes	13.86
numCountriesOfOrigin	moreCountriesOfOrigin	5.05

Table 2: Binarization of numeric features.

man’s correlation coefficient instead of Pearson in plotting the correlation heatmap in Fig 3.

We removed the following features because they were highly correlated to another feature: *awardNominationsExcludeWins* (0.86, correlated with *awardsAndNominations*), *castNumber* (0.74, with *totalCredits*). Since all features regarding reviews and *numVotes* were highly correlated with each other, we dropped them all with the exception of *numVotes*, which has the least skewed distribution, and *criticReviewsRatio*, which offer insights of a different nature. We however decided to conserve information about the presence/absence of reviews by users and critics with the creation of two new boolean features: *hasUserReviews* and *hasCriticsReviews*.

Moreover, we binarized features which overwhelmingly took on the value 0 (or 1, in the case of *numCountriesOfOrigin* and *directorsCredits*). The results of these transformations are reported in Table 2.

The total number of final features is 27; they are listed in Tables 3 and 4. Among the remaining numeric features, the heatmap in Fig. 4 shows that *externalLinks* exhibits a moderate to high positive correlation with *criticReviewsRatio* (0.61) and a low to moderate correlation with *numRegions*, *numVotes* and *totalImages* (around 0.47), while *numVotes* is also mildly correlated to *totalImages*, *totalCredits*, *criticReviewsRatio*, and *companiesNumber* (between 0.47 and 0.52).

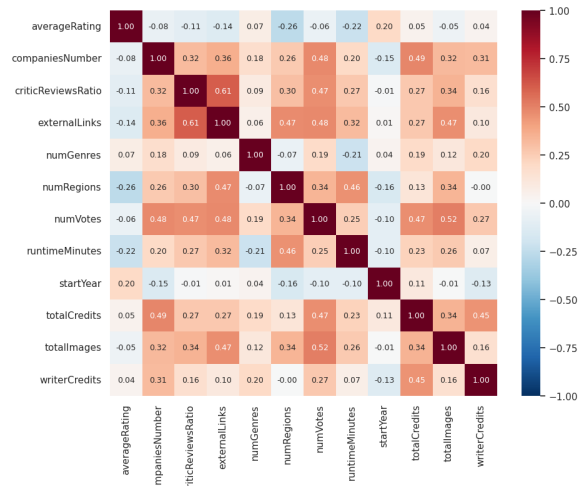


Figure 4: Correlation heatmap after attribute removal.

Feature	Description	Type
runtimeMinutes	Primary runtime of the title, in minutes.	Continuous. Ratio-scaled.
startYear	Represents the release year of a title. In the case of TV Series, it is the series' start year.	Discrete. Interval.
numVotes	Number of votes the title has received.	Discrete. Ratio-scaled. Log-transformed.
numRegions	The regions number for this version of the title.	Discrete. Ratio-scaled. Log-transformed.
totalImages	Total number of images for the title within the IMDb title page.	Discrete. Ratio-scaled. Log-transformed.
totalCredits	Total number of credits for the title.	Discrete. Ratio-scaled. Log-transformed.
criticReviewsRatio	The proportion of reviews from critics on the total number of reviews of a record. For unreviewed records, it is set to 0.	Continuous. Ratio-scaled.
numGenres	The number of genres listed for the variable <i>genres</i> .	Discrete. Ratio-scaled.
companiesNumber	Total number of companies that worked for the title.	Discrete. Ratio-scaled.
externalLinks	Total number of external links the title has within the IMDb page.	Discrete. Ratio-scaled.
writerCredits	Total number of writer credits of the title.	Discrete. Ratio-scaled.
averageRating	Weighted average of all the individual user ratings.	Continuous. Ratio-scaled.

Table 3: Numeric attributes after data preparation.

Feature	Description	Type
originalTitle	Original title, in the original language.	Categorical; mostly unique classes (i.e. names).
titleType	The type/format of the title (e.g. movie, short, tvseries, tvepisode, video, etc.).	Categorical, 10 classes.
countryOfOrigin	The country where the title was primarily produced.	Categorical, 153 classes. Some titles can belong to more than 1 class.
genres	The genre(s) associated with the title (e.g., drama, comedy, action).	Categorical, 29 classes. Some titles can belong to more than 1 class.
regions	The regions for this version of the title.	Categorical, 171 unique classes. Some titles can belong to more than 1 class.
rating	IMDb title rating class.	Ordinal, 10 values.
hasAwards	Whether the title has won at least an award	Asymmetric binary.
hasAwardsOrNominations	Whether the title was at least nominated for an award.	Asymmetric binary.
canHaveEpisodes	Whether the title can have episodes.	Asymmetric binary.
hasVideos	Whether the title IMDb title page has at least one video.	Asymmetric binary.
moreCountriesOfOrigin	Whether the title was produced in more than one country.	Asymmetric binary.
hasCriticsReviews	Whether the title has any critic reviews.	Asymmetric binary.
hasUserReviews	Whether the title has any user reviews.	Asymmetric binary.
moreDirectorsCredits	Whether the title has more than one director credits.	Asymmetric binary.
hasQuotes	Whether the title has any quotes within the IMDb page.	Asymmetric binary.

Table 4: Categorical, ordinal, and binary attributes after data preparation.

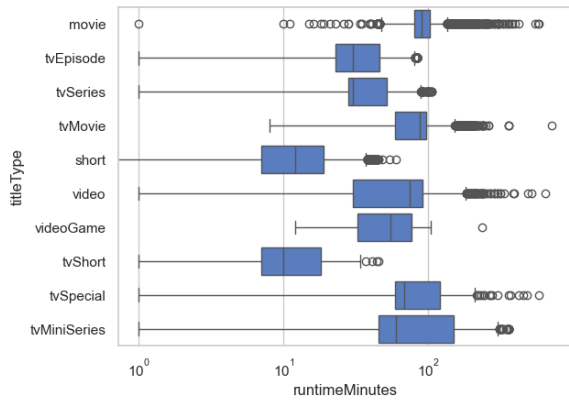


Figure 5: Distribution of runtimes across *titleTypes*.

1.6 Handling Missing Values

At this point, we still have missing values for *countryOfOrigin*, *region*, *genres*, *runtimeMinutes* and *writerCredits*; some were preexisting, others were removed as detailed Subsec. 1.4. Since the distribution of *runtimeMinutes* depends on *titleType* (cf. Fig 5), and the same holds true for *writerCredits*, we imputed their missing values using the median value within each respective class.

On the other hand, we chose not to replace missing values for *genres*, *region* or *countryOfOrigin*. They can easily be handled when performing the encoding for our methods without introducing bias, therefore we'd rather preserve the original distributions and correlations.

1.7 Clustering

As part of the data understanding phase, we conducted hierarchical clustering on the dataset, using all numeric attributes.

Since the size of the dataset made it unfeasible to perform hierarchical clustering directly because of the computational cost of computing distances between all records, we firstly reduced our dataset to a more manageable 10,000 records, running k-means clustering with $k=10,000$.

We then performed hierarchical clustering on the centroids of K-Means clusters, experimenting with different linkage methods and inspecting the resulting dendrograms. We found that using average linkage yielded the most satisfactory results. We then cut the dendrogram at an appropriate threshold and, after propagating cluster labels back to the entire dataset, analyzed the composition of the clusters.

We found 13 clusters of varying sizes, with the top cluster grouping one fifth of the entire dataset and the top four clusters grouping half of the

dataset. Five clusters contained less than 5,000 records each.

For reasons that will become apparent in 2.1, we concentrated our analysis on two smaller clusters: C1 (1,668 records) and C2 (3,923). These clusters belong to the same supercluster, that is merged latest of all with the rest of the dataset.

Both clusters are made up primarily from movies (with some tv series), and according to their distribution w.r.t. *numVotes*, probably contain blockbuster movies and very popular TV series 6. In addition to that, almost 57% of records from C1 received awards, a proportion that increases to 80% when considering also nominations. C2 records were also well received, with 45% of them having received or being considered for awards. In addition to that, 87% of records from C1 and 45% of records from C2 have videos on their IMDB page and almost all records from both C1 and C2 have both reviews from users and from critics.

2 ADVANCED DATA PRE-PROCESSING

2.1 Anomaly detection

2.2 Imbalanced Learning

In this section we will discuss several imbalance learning techniques in the context of binary classification. Our goal was to develop a system that is capable of predicting which titles have received any awards (as already seen in 2, only about 8% of our training records did). We can imagine that a similar system could be useful in a number of contexts, like for recommendation and showcasing purposes or for streaming platforms attempting to decide which titles to license when this information isn't already available (for example for newly released titles).

In our evaluation of different techniques, we decided to rely on the F1 score of the positive class, since we figured that from a business perspective it was equally important detecting as many award worthy titles as possible and avoiding false positives.

2.2.1 Resampling techniques

We established a performance baseline using a KNN classifier with no resampling techniques. We then experimented with three over-sampling strategies (Random Over-sampling,



Figure 6: Hierarchical clustering results on tabular data by runtime and number of votes.

SMOTE, ADASYN), two under-sampling strategies (Random Under-sampling and Edited Nearest Neighbors) and a combination of both (SMOTEEN). In all cases we resampled the dataset as to obtain a perfect balance between the positive and the negative class.

For each technique we fine tuned the hyperparameters of the classifier searching for the best F1 score: initially we performed random search on k s between 1 and 100, with euclidean and manhattan distance, both weighting neighbors and not. Since we noticed generally better performance with weighting of exemplars, we removed unweighted models from the grid search.

We also noticed that for over-sampling methods performance was rapidly decreasing as k got bigger, so we restricted our grid search to odd k s between 1 and 21. Since performance of under-sampling techniques wasn't degrading so rapidly, we extended the number of neighbors considered for these approaches to odd numbers between 1 and 41, but we noticed that scores were plateauing around 10, so we didn't consider it necessary to widen the search beyond that.

We then compared performance of different resampling strategies checking precision, recall and F1 score on 5-fold cross validation (cf fig 5).

Precious insights were also gathered by the inspection of the precision-recall curves (reported in fig 7) and average precision.

Over-sampling methods and the combination of SMOTE and Edited NN all show worse rank-

Method	precision	recall	f1
No strategy	.47 $\pm$.02	.31 $\pm$.01	.37 $\pm$.01
Random Over-saml.	.29 $\pm$.00	.62 $\pm$.01	.40 $\pm$.01
SMOTE	.29 $\pm$.00	.65 $\pm$.01	.40 $\pm$.01
ADASYN	.30 $\pm$.00	.59 $\pm$.01	.40 $\pm$.00
Random Under-saml.	.24 $\pm$.00	.86 $\pm$.01	.37 $\pm$.00
Edited NN	.50 $\pm$.01	.45 $\pm$.01	.47 $\pm$.01
SMOTE + Edited NN	.30 $\pm$.01	.68 $\pm$.01	.41 $\pm$.01
One Class SVM	.10 $\pm$.00	.62 $\pm$.01	.17 $\pm$.00

Table 5: Results of tested imbalanced learning methods on 5-fold cross validation.

ing capabilities than no resampling strategy at all. In particular these methods demonstrate much worse performance in the early retrieval range of operating points, meaning that they have much more false positives in the top ranked instances. This is also reflected in the lower precision scores that are not balanced by the higher recall. In addition to that, over-sampling techniques, even when relatively inexpensive at fitting time (Random Over-sampling), are slower at prediction time (because of the necessity to compute distances with all records in the over-sampled dataset) and have considerably higher space requirements.

Performance obtained with Random Under-sampling has similar F1 score to no resampling strategy, but shows better ranking capabilities, as demonstrated both by AP score and inspection of the PR curve.

The best results are obtained with Edited Nearest Neighbor. The PR curve for ENN almost

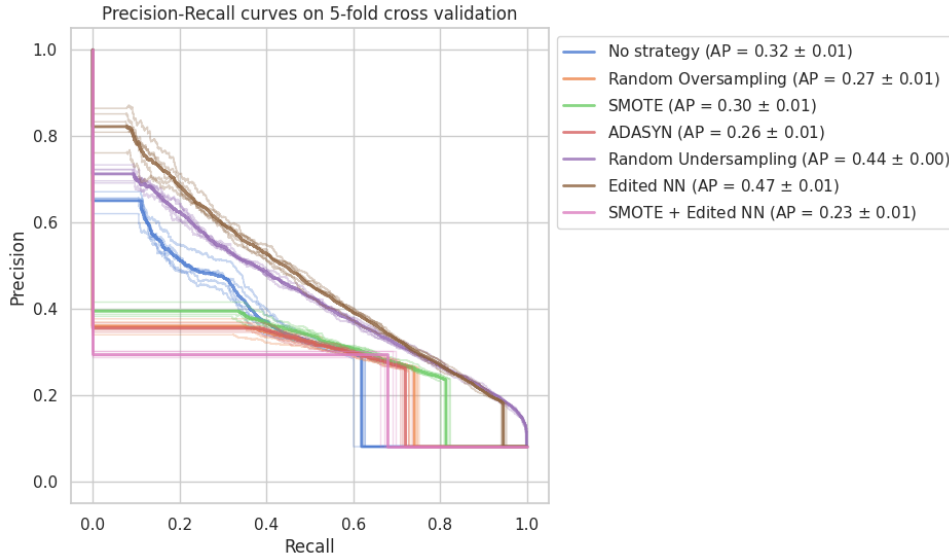


Figure 7: Precision-Recall curves on 5-fold cross-validation for different resampling methods

always dominates random under-sampling with minimal crossing-over for higher values of recall.

It seems to us that Random Under-sampling may be the preferred choice in some context in which fitting time would be a more pressing concern than the precision of the prediction, so we decided to assess both Edited NN and Random Under-sampling on the test set.

2.2.2 Decision threshold and assessment

Edited Nearest Neighbor showed satisfactory performance, improving by 0.1 the F1 score of the vanilla KNN and having the best precision among all tested methods (0.50) and recall at least better than the vanilla KNN. Nevertheless, we decided to post-tune the decision threshold of both our best performing models before assessment on the test set. For the pipeline using Random Under-sampling this resulted in setting a higher threshold (0.77), while for Edited NN the threshold had to be lowered to 0.38. Fig 8 illustrates the effect of changing the decision threshold for the pipeline using Random Under-sampling: we can observe that at the default threshold the model would have better recall but much worse precision than at the tuned threshold.

We finally assessed both models on the test set, obtaining very similar results: ENN obtained a F1 score of 0.49, while Random Under-sampling 0.47; inspecting the other metrics we can confirm that the worse performance of Random Under-sampling is entirely due to a lower precision (0.42 vs 0.44 for ENN), as we had expected from the inspection of the PR curves in validation.

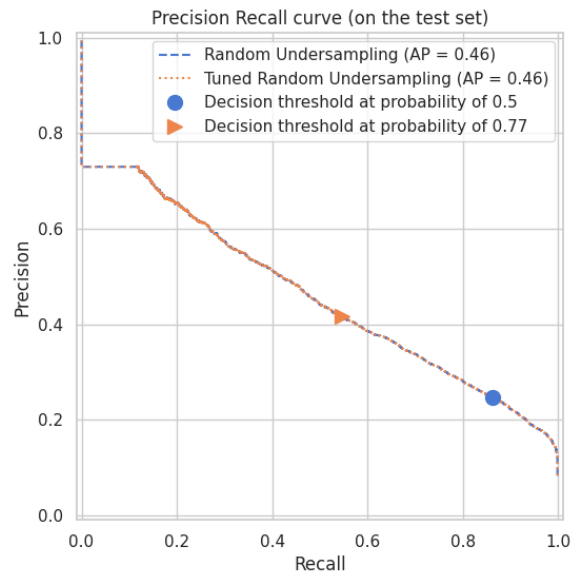


Figure 8: Decision threshold for KNN with Random Under-sampling before and after tuning.

2.2.3 Exploration of alternative techniques

As we detailed in 2.1, records that have received awards and nominations were often deemed anomalous. We decided to leverage this observation adopting a classification based anomaly detection technique for our classification task.

We trained a One Class SVM with RBF kernel without hyperparameter tuning on records that didn't receive any awards in order to learn a discriminative boundary around normal instances. Table 5 show results obtained by this method on 5-fold cross validation. We ultimately deemed the performance of this method unsatisfactory because it produced too many false positives.

3 ADVANCED CLASSIFICATION

In this section, we will examine the performance of various advanced classification methods w.r.t two multiclass classification tasks, aiming to predict the *rating* class and the *titleType* of our records.

Since no features showed high correlation with *averageRating*, while some did exhibit different distribution tendencies w.r.t. *titleType*, we anticipated that the second classification task would yield better results.

In evaluating and comparing our classification models, we selected macro F1 as the main performance metric, given that the different target classes have variable support. For each model, we additionally plotted the Precision-Recall (PR) curves for each class using the one-vs-rest strategy and reported its PR AUC. We favored PR curves instead of ROC curves because they are more informative in multiclass and imbalanced settings.

3.1 Pre-processing

For each model, we built a pipeline which proceeded to first preprocess the data (we used the training and test set, cleaned and prepared as detailed in Sec. 1) and then pass it to the model, in order for it to be fitted on the training set or applied on the test set.

The preprocessing step included scaling the numerical features, if required by the method, and one-hot encoding the categorical variables. For both tasks, we of course dropped the target variable, including *averageRating* for the rating task.

For *countryOfOrigin* and *region*, we only kept the 25 most frequent countries, aggregating the rest into an *Other* category, to reduce dimensionality. For the MultiLayerPerceptron Classifier (and, later, for the regression task), we further reduced dimensionality, keeping only the top nine countries for both features and dropping *hasAwards* on the account of its positive correlation to *hasAwardsOrNominations*, in order to further reduce computational costs and training time during hyperparameter tuning.

3.2 Logistic Regression

As a baseline, we first tried a Logistic Regression model. As the algorithm assumes that the features follow a Gaussian distribution, we scaled numerical features using PowerTransformer, which reduces skewness and stabilizes variance.

3.2.1 Rating

Using the default parameters yielded a macro F1 of 0.144 and an accuracy of 0.388. We performed a grid search using 5-fold cross validation, which resulted in a very small increase, with a 0.146 average macro F1 on the validation sets and 0.152 on the training set, which at least proves no overfitting. We tested the following hyperparameters (the values for the final model are in bold):

```
{ 'penalty': ['l2', 'l1', 'elasticnet'],
  'C': [0.001, 0.01, 0.1, 1, 10, 100],
  'l1_ratio': [0.1, 0.3, 0.5, 0.7, 0.9],
  'solver': ['lbfgs', 'sag', 'saga', 'newton-cg',
             'newton-cholesky', 'liblinear'] }
```

On the test set, the final model achieved a **macro F1** of **0.145** and an **accuracy** of **0.388**, comparable to the default model. The poor performances across decision thresholds are confirmed by the **PR AUC** of only **0.193**.

The model has managed to generalize some information only for the classes with the highest support: the performances peak for the mode, (7,8] (F1: 0.544), its neighboring classes are confused for it most of the time, and as we move to the tail of the bell distribution, it achieves consistently poor results (F1 < 0.05).

Looking at the model's coefficients, the overall most important feature is *hasCriticReviews* (mean absolute value > 1.2, while for the other coefficients in the top 15 is in the interval [0.6, 0.7]): it is the first or second-biggest coefficient for the best performing classes (i.e. for the interval (5,8]), and for all of them it takes on a negative value, i.e. having critical reviews lowers the probability of a record of having an 'average' rating.

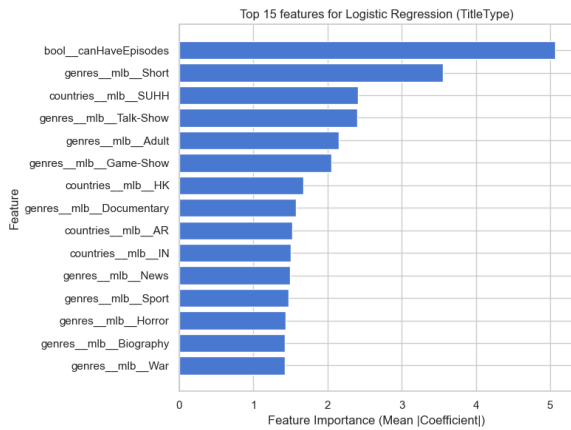


Figure 9: Top 15 most important features for *titleType* (LR).

3.2.2 Title Type

For this task, the final model resulting from the grid search achieved an average validation macro F1 of 0.653 (on the training set: 0.664). The final parameters chosen were: 'C': 100, 'penalty': 'l2', 'solver': 'newton-cg'.

On the test set, the model achieved a **macro F1** of **0.649** and an accuracy of **0.908**; the improvement w.r.t. the default parameters (0.647 and 0.908) is, again, quite small. The **PR AUC** is **0.704**; the curve for each class reflects its performance.

Given that both tasks have 10 target classes, our intuition about *titleType* yielding better results is confirmed.

The four most popular classes (*tvEpisode*, *short*, *movie*, *tvSeries*) achieve great results, with F1 > 0.9. The best performing and most popular class, *tvEpisode*, peaks with 0.978. All other classes (except *tvShort*) achieve a F1 score > 0.4, which is still significantly more than random chance. *Videogame*, in spite of its low support, manages to achieve very good results (F1 0.76).

The 15 most important features are reported in Fig. 9. Overall, the two most informative features are:

- *canHaveEpisodes*: it has very high positive values for *tvSeries* (+13.15) and *tvMiniSeries* (+12.18) and negative for all other classes; it's also the most important feature for *movie* (-4.75).
- *genre Short*: of course, it's the most important feature for *short* and *tvShort*, and it's the second most important for *video* with positive values; it's high up in the ranking with a negative value for *movie*, *tvEpisode*, *tvMovie*.

3.3 Linear Support Vector Machine

For Support Vector Machines (SVMs), we scaled numerical features using *RobustScaler*, as our data contains outliers. We tested the linear kernel and the Radial Basis Function kernel separately to exploit *sklearn*'s optimized *LinearSVC* implementation.

3.3.1 Rating

For *LinearSVC*, we tested the following parameter grid using a grid search with 5-fold cross validation:

```
{ 'C': [0.001, 0.01, 0.1, 1, 10, 100],
  'penalty': ['l1', 'l2'],
  'loss': ['squared_hinge', 'hinge'],
  'max_iter': [5000],
  'class_weight': [None, 'balanced'] }
```

It resulted in an average macro F1 of 0.158 for validation and of 0.164 on the training set.

On the test set, w.r.t. Logistic Regression, the model achieved a higher **macro F1 (0.158)**, but a lower **accuracy (0.270)**. Indeed, the F1 for the best performing classes has decreased ((7,8]: 0.496 vs 0.544; (6,7]: 0.215 vs 0.356, (5,6]: 0.259 vs 0.283), in favor of slightly better performances for classes with lower support: the decrease in accuracy is linked to the fact that the model tries to predict these lower support classes more often, rather than focusing merely on the central classes. This led to a notable increase in F1 for (3,4] (0.113 vs 0.04) and (9,10] (0.145 vs 0.046), but it is clear that the model still struggles. The **PR AUC** is also slightly worse, **0.175**.

The confusion matrix confirms this tendency: if for Logistic Regression predictions were concentrated on the best performing classes, with many zeros populating the lowest classes, now confusion is spread throughout.

The most important feature is *tvEpisode* (mean absolute value: 0.175): it is at number one, with negative values, for classes from (2,3] to (5,6] — this would suggest that serialized content is less likely to have bad rating. Classes (7,8] and (8,9] both have *movie* as most important feature, with negative contributions (-0.207 and -190, respectively). This might explain why the latter is mostly classified as the former, which is the class with the highest support.

3.3.2 Title Type

The model returned by the grid search yielded an average validation macro F1 of 0.646 (vs 0.654 on the training set); the parameters were

```
'C': 0.1, 'class_weight': 'balanced',
'loss': 'squared_hinge', 'max_iter':
5000, 'penalty': 'l1'.
```

On the test set, the results were slightly lower than Logistic Regression: **0.640 macro F1**, **0.889 accuracy**, **0.674 PR AUC**. Indeed, performances for almost all classes have decreased to some degree, with a remarkable dip in precision for *videogame* (0.486 vs 0.799).

The overall most important features are the same (*canHaveEpisodes* and genre *Short*), but now only the first stands out (mean absolute value close to 2.5), while *Short* has a similar value to the rest of the top features, < 1.

Unexpectedly, the most important feature for the *titleType short* is now *canHaveEpisodes*, with quite an 'advantage' (coef: -2.18, vs genre *Short* 1.43).

3.4 Support Vector Machine with Radial Basis Function (RBF)

The power of SVMs comes at a high computational cost, which is why we only tested one kernel function (RBF) and why we opted for an 80/20 holdout, instead of a k-fold cross-validation, to carry out hyperparameter tuning. We also defined a small hyperparameter grid:

```
{ 'C': [0.1, 1, 10],
  'gamma': ['scale', 0.01, 0.1],
  'class_weight': [None, 'balanced'] }
```

3.4.1 Rating

The best model returned by the grid search had the following hyperparameters: 'C': 10, 'gamma': 0.1, 'class_weight': None. It achieved a macro F1 of 0.265 on the validation set and of 0.698 on the training set, suggesting overfitting. We therefore reran the grid search, excluding the highest values for C and gamma in order to increase regularization. The best performing model was less overfitted, but also achieved lower results (macro F1 0.206 for validation and 0.316 for training). Its parameters are: 'C': 1, 'class_weight': 'balanced', 'gamma': 'scale'.

On the test set, the overfitted model proved to be the best one, achieving a **macro F1** of **0.257** (vs 0.219) and an **accuracy** of **0.431** (vs 0.314). The resulting **PR AUC** is **0.231**. This model is also the best performing one thus far, with improvements observed across almost all classes.

Looking at the confusion matrix, we can see that the model doesn't completely snub lower

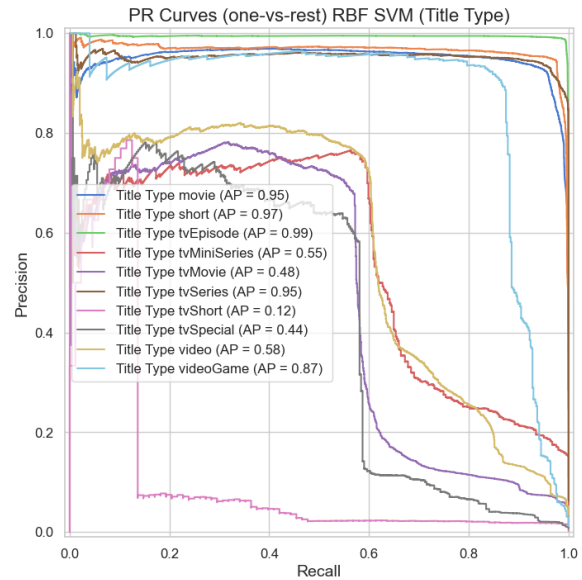


Figure 10: PR curves for *titleType* (SVM with RBF).

support classes, like Logistic Regression did, but is more accurate than LinearSVC. Especially for the more popular (and better performing) classes, confusion is mostly among neighboring classes.

3.4.2 Title Type

Similarly to what had happened with *rating*, the first grid search returned an overfitted model (macro F1 for validation: 0.746, vs for training: 0.905). Forcing regularization resulted in less overfitting, but also in a much poorer performance (validation macro F1: 0.206, train macro F1: 0.316); for this reason, we chose the first classifier. Its parameters are: 'C': 10, 'class_weight': None, 'gamma': 'scale'.

On the test set, this model clears all previous attempts, with a **macro F1** of **0.746** and an **accuracy** of **0.938**.

The performances for all classes increased. *tvEpisode* now achieves an F1 of 0.99; *videogame* improved greatly, reaching 0.884 F1. Nevertheless, we only get a **PR AUC** of **0.689**; indeed, the curves in Fig. 10 show how the precision for half of the classes sinks at around 60% recall. This means that while the model shows excellent performance metrics on a per-class basis, it has difficulty maintaining a reliable performance when attempting to comprehensively identify instances of minority classes.

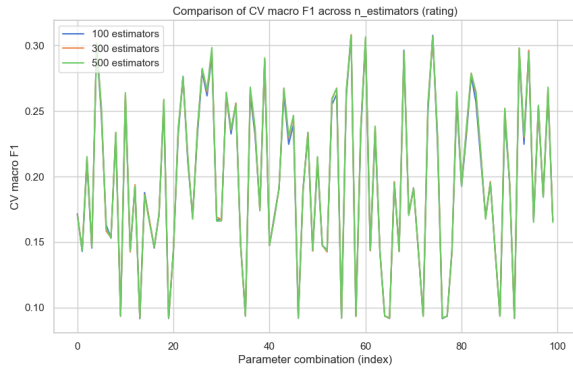


Figure 11: Avg cross-validation macro F1 for different parameter configurations across varying numbers of estimators for *rating* (RF).

3.5 Random Forest

Since this method is based on trees, no scaling is needed for numerical features. To cut computational costs, we used 3-fold cross validation during hyperparameter tuning.

Indeed, there are many possible hyperparameters to tune. For each task, we ran three Randomized Searches, for a varying number of estimators — 100, 300 and 500, respectively. For each, we tested the same 100 random combinations of parameters from the following grid:

```
{ 'criterion': ["gini", "entropy"],
  'max_depth': [None, 5, 15, 25],
  'min_samples_split': [2, 5, 10],
  'min_samples_leaf': [1, 5, 10],
  'class_weight': [None, 'balanced'] }
```

3.5.1 Rating

Before proceeding with hyperparameter tuning, we first tested a Random Forest using default parameters, which yielded an average validation macro F1 of 0.296, and an even better 0.302 macro F1 score on the test set.

As shown in Fig. 11, the number of estimators is not a strong driver of performance. Although the performance of the other two models was very close, we chose the best model with 300 base classifiers, which yielded an average macro F1 equal to 0.308 for validation, and equal to 0.968 for train. Its parameters are: `'n_estimators': 300`, `'min_samples_split': 5`, `'min_samples_leaf': 1`, `'max_depth': None`, `'criterion': 'gini'`, `'class_weight': 'balanced'`.

Whether such a gap between validation and train macro F1 should be seen as a sign of overfitting for Random Forest is debated in the literature: it has been argued that, even if the single trees can

overfit, they tend to do so in different ways, and that combining them in an ensemble reduces the overall variance. Thus, although pruning the individual trees can be beneficial, unless the dataset is very simple, we could consider Random Forest as particularly robust against overfitting.

The model's performance on the test set increases, making it the best model thus far: it achieves a **macro F1** of **0.325**, an **accuracy** of **0.452**, and a **PR AUC** of **0.318**.

It achieves the most balanced performance, with six classes out of ten scoring an $F1 \geq 0.3$. Confusion remains spread out for the minority classes (from 0 to 4), whereas for the others it remains concentrated amongst neighboring classes.

The Mean Decrease in Impurity (MDI) returned as the most important features *numVotes* and *startYear* (almost 0.10), followed by *totalCredits* (around 0.9).

3.5.2 Title Type

The default model achieved an average macro F1 of 0.707 on validation and a score of 0.728 on the test set.

The performances for the different parameters combinations tested during the Random Searches were again very similar, with the configurations using 100 estimators yielding marginally lower scores. Indeed, the three best models shared the same hyperparameter values. We chose the configuration with 300 trees: the validation performances were only marginally better than that of the 500-trees model, but we also favored simplicity.

The model yielded 0.767 and 0.975 for validation and training, respectively. Its hyperparameter configuration is the following: `'n_estimators': 300`, `'min_samples_split': 10`, `'min_samples_leaf': 1`, `'max_depth': None`, `'criterion': 'entropy'`, `'class_weight': 'balanced'`.

On the test set, it clears previous models, achieving **0.777 macro F1** and **0.940 accuracy**. Half of the classes yield $F1 > 0.95$. It generalizes better on the minority classes *tvMovie* and *tvMiniSeries*: although there is still confusion with the more popular *movie* and *tvSeries*, they are correctly classified most of the time (cf. the confusion matrix in Fig. 13).

This model's superiority across decision thresholds is supported by the PR curves in Fig. 12, and further quantified by its **PR AUC** of **0.818**.

Checking the 15 most important features (Fig 14), we see how the two most important features for the Logistic Regression model, *canHaveEpisodes*

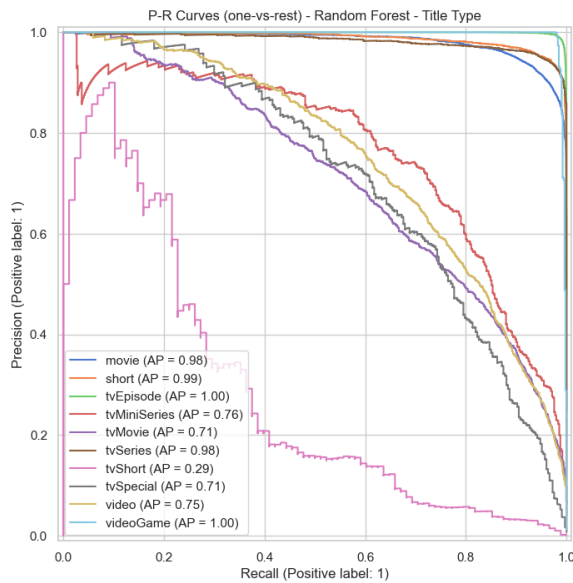


Figure 12: PR curves for *titleType* (RF).

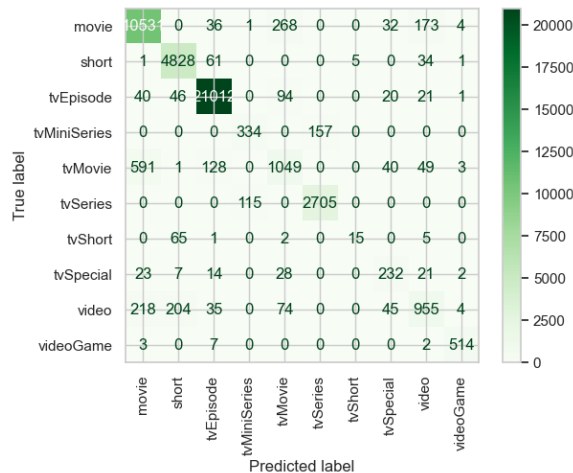


Figure 13: Confusion matrix for *titleType* (RF).

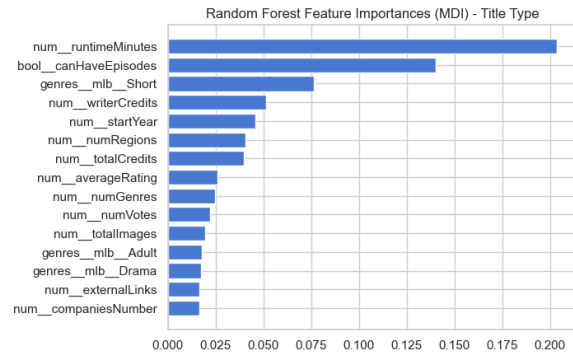


Figure 14: Top 15 most important features for *titleType* (RF).

and the genre *Short*, have been overtaken by *runtimeMinutes*. Indeed, we had uncovered a relationship between *runtimeMinutes* and *titleType* in Subsec. 1.6.

Finally, it's worth pointing out that prior experimentation keeping the default hyperparameter 'class_weight': None instead of 'balanced' lead to a best model scoring a significantly lower macro F1 score of 0.728. This is indicative of the importance of this hyperparameter for this unbalanced task.

3.6 LightGBM

Within the Gradient Boosting family of algorithms, we selected LightGBM for its efficiency. Since it exploits Decision Trees as base classifiers, numerical features don't require scaling.

Similarly to what we did for Random Forests, we performed hyperparameter using a Randomized Search with 100 iterations and a 3-fold cross-validation to test the same random combinations with 100 and 300 estimators (we excluded 500 estimators because it resulted too computationally expensive). The hyperparameter configurations were sampled from the following grid:

```
{ 'boosting_type': ['gbdt', 'goss', 'dart'],
  'num_leaves': np.arange(20, 100, 10),
  'max_depth': [-1] + list(np.arange(3, 13)),
  'learning_rate': [0.01, 0.03, 0.05, 0.1],
  'reg_alpha': [0, 0.01, 0.1, 1],
  'reg_lambda': [0, 0.01, 0.1, 1] }
```

3.6.1 Rating

The best resulting model has the following hyperparameters: 'n_estimators': 300, 'reg_lambda': 0.1, 'reg_alpha': 0.01, 'num_leaves': 50, 'max_depth': 10, 'learning_rate': 0.1, 'boosting_type': 'gbdt'. It yielded an average cross-validation



Figure 15: Avg. cross-validation macro F1 for different parameter configurations for 100 and 300 DTs for *rating* (LightGBM).

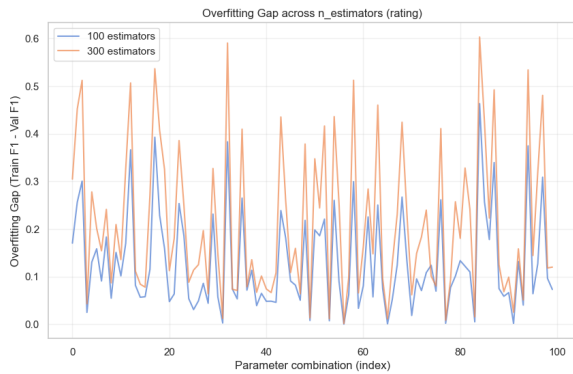


Figure 16: Overfitting gap for different parameter configurations for 100 and 300 DTs for *rating* (LightGBM).

macro F1 of 0.285 and of 0.819 on the training set, showing signs of overfitting.

Given the same hyperparameter configuration, increasing the number of estimators leads to a better performance (Fig. 15), but also to a larger overfitting gap (i.e. the difference between training and validation macro F1) (Fig. 16).

Fig. 17 shows how the overfitting gap grows exponentially as performances increase, suggesting that the data is hard to model.

On the test set, the model achieved **0.289 macro F1**, **0.440 accuracy**, and **0.299 PR AUC**, positioning itself at second place, under Random Forest.

Consistently with Random Forest, *startYear* and *numVotes* emerged as the most important features, with comparable gain values.

3.6.2 Title Type

Analyzing the results of the Random Searches leads to the same discovery we had for *rating*: given a certain hyperparameter configuration, the average cross-validation macro F1 grows for 300 estimators w.r.t. 100, but so does the overfitting

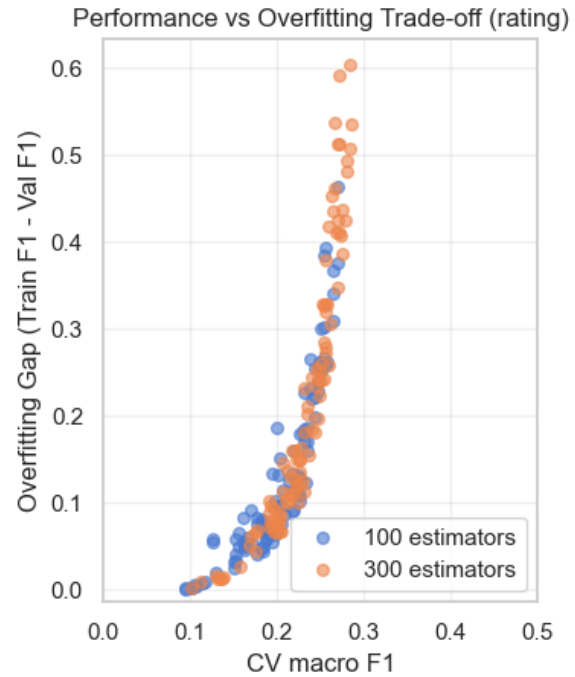


Figure 17: Trade-off between avg validation macro F1 and overfitting gap for *rating* (LightGBM).

gap. Nevertheless, Fig. 18 shows how the relationship between the overfitting gap and the validation macro F1 grows more slowly. Moreover, the best performing models are only moderately overfitted (gap of around 0.2 vs 0.6 for *rating*.)

The best model yields a cross-validation macro F1 of 0.791 against a training macro F1 of 0.982. Its parameters are: 'n_estimators': 300, 'reg_lambda': 1, 'reg_alpha': 0, 'num_leaves': 20, 'max_depth': 10, 'learning_rate': 0.1, 'boosting_type': 'gbdt'.

On the test set, it reached **0.795 macro F1**, **0.949 accuracy**, and **0.847 PR AUC**, surpassing Random Forest. Performances for almost all classes improved, with *tvEpisode* reaching 0.99 F1. The lowest F1 score, excluding *tvShort*, is 0.688 for *tvMovie*. The PR curves in Fig. 19 also highlight the improvement for the minority class *tvSpecial*, which indeed scored 0.747 F1 (vs 0.667 for Random Forest).

Looking at the feature importance (Fig. 20), the top three most important features are the same as for Random forest, except that genre *Short* is slightly more important than *canHaveEpisodes*, and that *runtimeMinutes* by far the most significant.

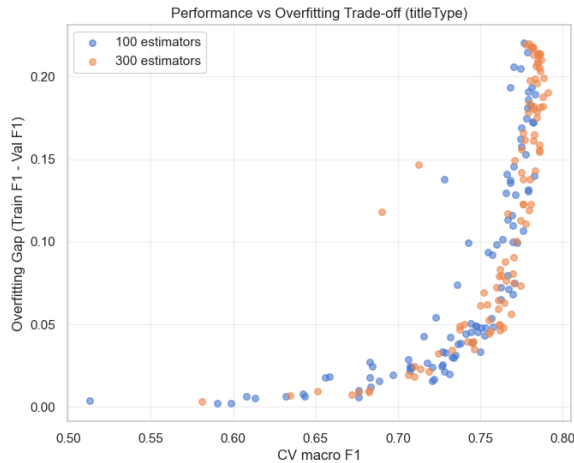


Figure 18: Trade-off between avg validation macro F1 and overfitting gap for *titleType* (LightGBM).

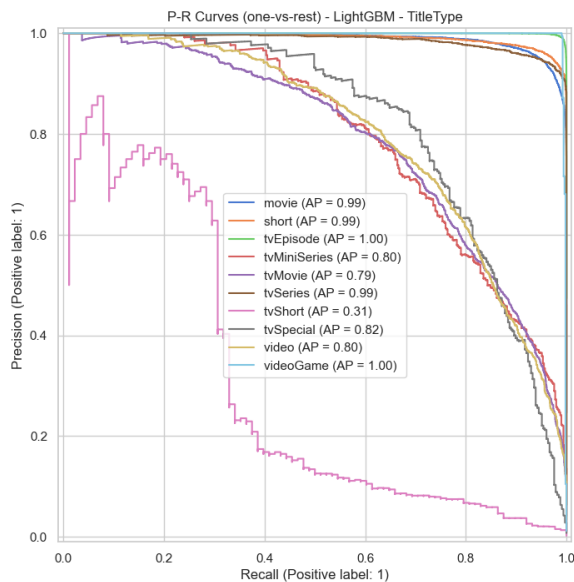


Figure 19: PR curves for *titleType* (LightGBM).

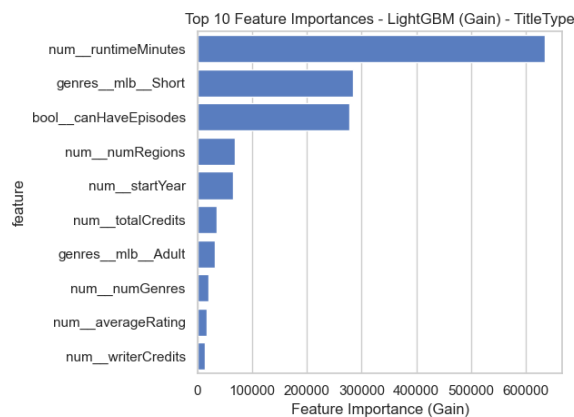


Figure 20: Top 10 most important features for *titleType* (LightGBM).

3.7 Multilayer Perceptron

As anticipated in Subsec. 3.1, for this model, we reduced the dimensionality of the dataset to make training faster. We also scaled numerical features using RobustScaler.

Since a comprehensive grid search for hyperparameter tuning would have been too computationally expensive, we ran three Randomized Searches using 80/20 holdout and 100 iterations for a varying number of hidden layers, ranging from one to three. As for the size of hidden layers, we tested the following architectures, varying the number of hidden units and the decreasing patterns:

```
[(32,), (64,), (128,)],
[(64,32), (128,64), (128,32)],
[(128,64,32), (64,64,32)]
```

The rest of the parameters are sampled from the following grid:

```
{ 'activation': ['relu', 'tanh'],
  'alpha': [0.0001, 0.001, 0.01],
  'learning_rate_init': [0.0001, 0.001, 0.01] }
```

For both tasks, the MLPClassifier positioned itself in the lower half of the ranking. Further exploration of more advanced Deep Learning architectures is needed to better leverage the potential of Neural Networks.

3.7.1 Rating

We first tried using the default parameters, which yielded 0.239 macro F1 on the validation set and 0.316 on the training set, whereas on the test set it achieved 0.240 macro F1 and 0.415 accuracy.

After hyperparameter tuning, the best model ended up having two hidden layers, yielding 0.262 validation macro F1 and 0.387 for training. Its parameters are: 'hidden_layer_sizes': (128, 64), 'learning_rate_init': 0.001, 'alpha': 0.01, 'activation': 'relu'.

Plotting the trade-off between the overfitting gap and the validation macro F1 (Fig. 21) shows that the gap tends to increase as performance increases, but the relationship is not as clear as it was for LightGBM. Moreover, most underfitted models only have one layer.

On the test set, the model achieved a macro f1 of 0.244, an accuracy of 0.413 (lower than the default model), and a PR AUC of 0.244.

We also tried implementing early stopping: the macro F1 sank to 0.236, but the accuracy increased slightly, reaching 0.425.

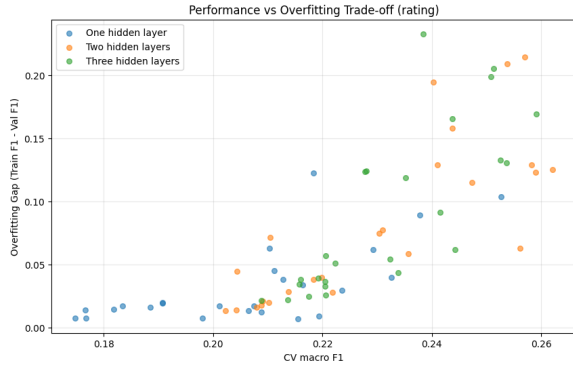


Figure 21: Trade-off between validation macro F1 and overfitting gap for *rating* (MLP).

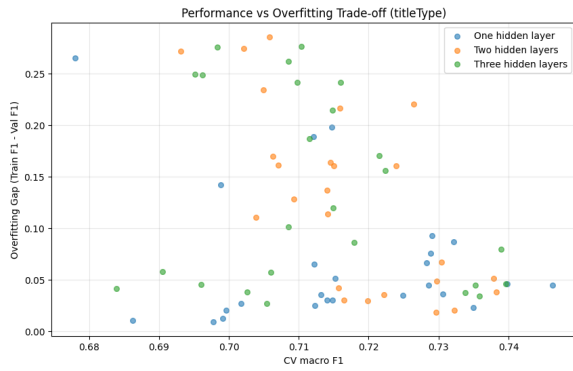


Figure 22: Trade-off between validation macro F1 and overfitting gap for *titleType* (MLP).

3.7.2 TitleType

In this case, in terms of macro F1 the default model reaches 0.729 on the validation set, 0.857 on the training set, and 0.730 on the test set. It also achieves 0.931 test accuracy.

After hyperparameter tuning, the best model turns out to be relatively simple, with just one layer and 64 hidden units. It yields a macro F1 of 0.746 on validation and 0.791 on the training set, excluding overfitting. Its hyperparameters are: `'alpha': 0.01`, `'hidden_layer_sizes': (64,)`, `'learning_rate_init': 0.001`, `'activation': 'tanh'`.

Indeed, the scatterplot in Fig. 22 reveals how more layers do not necessarily lead to a better performance, but they do lead to a tendency to overfit.

On the test set, the model still achieves **0.738 macro F1**, coupled with **0.934 accuracy** and **0.784 PR AUC**. Implementing early stopping leads to a drop in macro F1 to 0.728, as it performs slightly worse on minority classes.

Model	Macro F1	Accuracy	PR AUC
RF	0.325	0.452	0.318
LightGBM	0.289	0.440	0.299
SVM w/ RBF	0.257	0.431	0.231
MLP	0.244	0.413	0.244
LinearSVM	0.158	0.270	0.175
LR	0.145	0.388	0.193

Table 6: Performance comparison of different models on *rating*.

Model	Macro F1	Accuracy	PR AUC
LightGBM	0.795	0.949	0.847
RF	0.777	0.940	0.818
SVM w/ RBF	0.746	0.938	0.689
MLP	0.738	0.934	0.784
LR	0.649	0.908	0.704
LinearSVM	0.640	0.889	0.674

Table 7: Performance comparison of different models on *titleType*.

3.8 Conclusions

The performances of the different models for the two tasks are summarized in Table 6 and 7.

As anticipated, the task of predicting *titleType* yielded better results across models, which, just like for DM1, is likely due to better class separability and stronger feature association.

All models achieved great results for the more popular classes, with an F1 around 0.9, while the results for the minority classes varied from around 0.4 to a robust 0.7 for the best models. In particular, the best models achieved great results for *videogame*, and were better at separating *tvMovie* and *tvMiniSeries* from their more popular counterparts, *movie* and *tvSeries*. Ensemble methods have clearly outperformed the rest.

With regard to *rating*, the performance of every model reflects the class distribution, which resembles a left-skewed Gaussian: performances peak for class (7,8], the mode, and quickly degrade. The models do not have a chance to generalize on the lowest and the highest classes, especially those in the 'tail', due to lack of representation. The best models, however, managed to concentrate their confusion on neighboring classes, especially around the peak of the distribution.

Finally, there have been some challenges tied to overfitting; future work could explore undersampling and oversampling techniques to improve generalization.

4 EXPLAINABILITY: SHAP FOR RANDOM FOREST CLASSIFICATION

5 ADVANCED REGRESSION

In this section, we will discuss the performance of two advanced regression methods in predicting the *averageRating*.

We used the same preprocessing pipeline detailed in Subsec. 3.1 and the feature set we had used for the Multilayer Perceptron.

We evaluated performances using the Determination Coefficient (R^2) and the Mean Squared Error (MSE). To set a baseline, we used a DummyRegressor which always predicts the mean of the target values observed in the training set, which yielded a R^2 of 0 and a MSE of 1.819 (equal to the variance of the target variable).

5.1 Random Forest Regressor

5.2 XGBoost Regressor

6 TIME SERIES ANALYSIS

6.1 Data Understanding and Preparation

This dataset contains data on the domestic (US and Canada) daily gross income at the box office for 1,134 movies, measured from their release date up to 100 days after release. The dataset was extracted from Box Office Mojo¹ and contains no missing values, because shorter time series were padded using noise augmented mean. Each time series was associated with the movie IMDB id, genre and rating (both continuous average rating and a binned version). In addition to the provided data, we used the IMDB free api² to retrieve some additional information about the titles represented in the dataset: title and release year of the movie, runtime (in minutes) number of votes, countries of origin, number of awards and nominations. We additionally aggregated the box office data as a rough estimate of the total gross income of the movie.

We then splitted our initial dataset reserving 25% of records for testing and conducted the rest of our analysis exclusively on the training set, as to prevent data leakage.

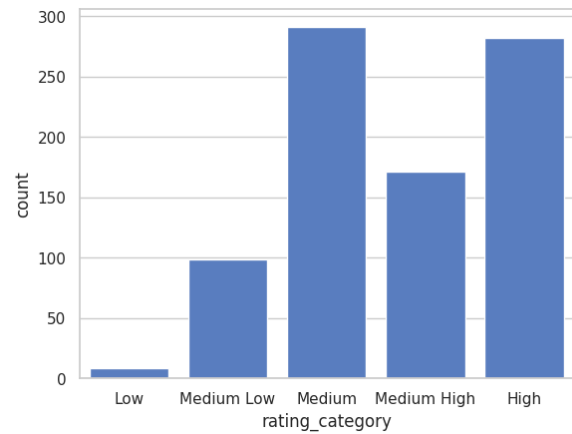


Figure 23: Distribution of rating category in the Time Series dataset.

6.1.1 Univariate analysis of variables

The distribution of number of votes, awards, nominations and total gross income show all a pronounced right skew, which is not unexpected. 90% of records had received awards or nominations. All records display normal runtimes for movies (between 90 and 200 minutes), except for 3 documentaries which were shorter than 45 minutes. We were able to detect some outliers also w.r.t. the release year: only two titles were released before 2010 (one in 1969 and one in 1984).

As for the distribution with respect to genres, any given movie can have more than one genre and the most popular genre was *Drama* (48%), followed by *Comedy* (36%), *Action* and *Adventure* (each of these 32%).

We also inspected the distribution of the binned version of rating, finding a considerable unbalance among classes, with *Medium* and *High* being the most frequent ones and the *Low* class representing less than 1% of titles (cf Fig. 23).

6.1.2 Multivariate analysis of variables

Figure 24 reports the Spearman correlation coefficient for pairs of numeric features. Besides obviously being correlated with each other, both awards and nominations are positively correlated with both the number of votes and rating. We also observe a moderate to strong positive correlation between the estimated gross income and number of votes. Perhaps surprisingly, total gross income is only weakly correlated with rating.

Positive correlation of the number votes with rating is something we didn't observe in the IMDB dataset, were, on the contrary these two features were weakly negatively correlated. We think this may be due to the fact that movies have on average

¹ <https://www.boxofficemojo.com/>

² <https://imdbapi.dev/>

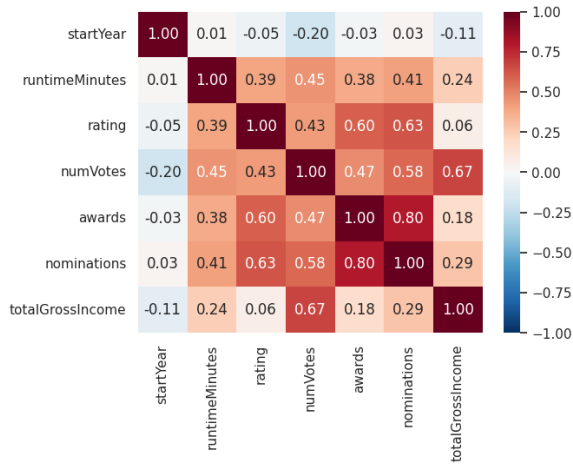


Figure 24: Correlation of numeric features.

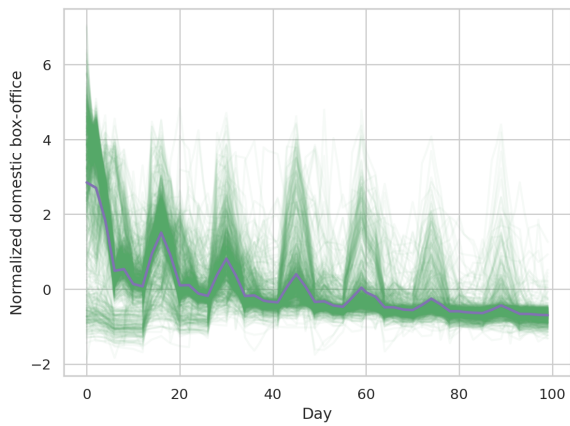


Figure 25: Time series (and mean time series, in purple) after amplitude scaling

lower ratings than TV Episodes, which, in turn, have fewer votes than movies. Since the current dataset is limited to movies, we do not observe this (spurious) relationship anymore.

6.1.3 Visualization

We conclude our understanding visualizing the signals for all time series after amplitude scaling (fig. 25). Even from this simple visualization we are able to draw some interesting insights.

First of all our time series in general show slightly negative trend and some seasonality with a cycle over around 14 timestamps: this observation was confirmed when we extracted Catch22 features from the time series and were able to verify that most of them had a periodicity of 13 or 14. This finding, which we initially took at face value, was somewhat puzzling, because we expected a day of week seasonality, with peaks in the weekends, while it made no sense for peaks to occur every other week. Surely enough, when

we finally checked against the original data on Box Office Mojo, we were able to confirm that the observed peaks are, in fact, the effect of day of week seasonality, and that the time series were altered before we received them, either during the extraction or during the preprocessing.

In addition to this discovery, we also were able to observe that, while most movies have profits well over the mean in the first several days after release, some others got a slow start at the box office. We were able to identify 145 movies that performed below average in the first day of box office. While this is a very coarse distinction, which incurs in some false positives, it was later confirmed by clustering. In addition to lower box office profits in the first days after release, these ‘late bloomers’ show no clear linear trend (in opposition to the negative trend observed for other movies) and tend to peak on the third weekend after release. We also observed that these were not, for the most part (93%), action movies.

6.2 Clustering

In our clustering analysis of time series we adopted two different approaches to the measure of similarity: shape-based similarity and structural similarity.

For the shape-based approach, we first subjected the time series to amplitude scaling and then measured the distances between records employing DTW as to minimize the impact of misalignment between time series. Since our time series were very short, we didn’t deem it necessary to apply any approximation method to the original signal; we did, however impose constraints (Sakoe-Chiba Band with radius 0.1) on the warping in order to reduce computation time and avoid pathological warpings.

For the global structural approach, we extracted mean, standard deviation and Catch22 features from the raw signals (without amplitude scaling) and then computed Euclidean Distance on this new feature matrix (which was normalized using a Standard Scaler).

6.2.1 Shape-based clustering

Since in our understanding phase we were able to recognize two distinct patterns, we first of all run K-Means with $k=2$ and obtained results very similar to those previously discussed. This clustering obtained a silhouette score of 0.85 and was able to identify 159 ‘late bloomers’.

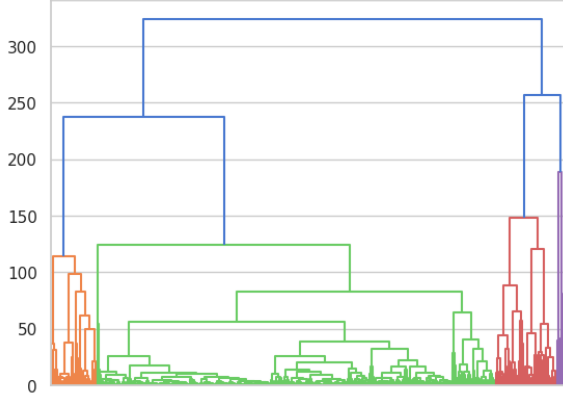


Figure 26: Dendrogram for shape-based hierarchical clustering.

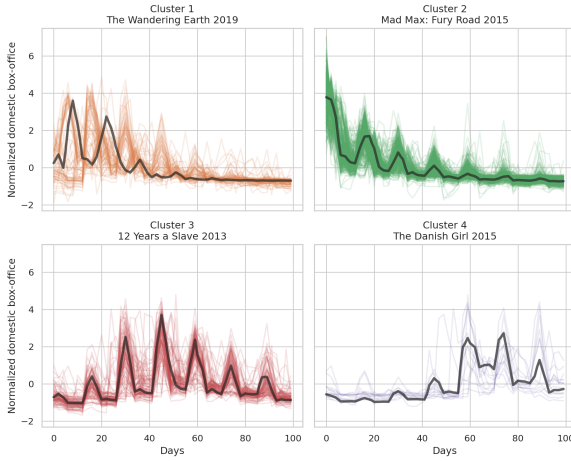


Figure 27: Clusters obtained by hierarchical clustering and respective cluster medoids (in black).

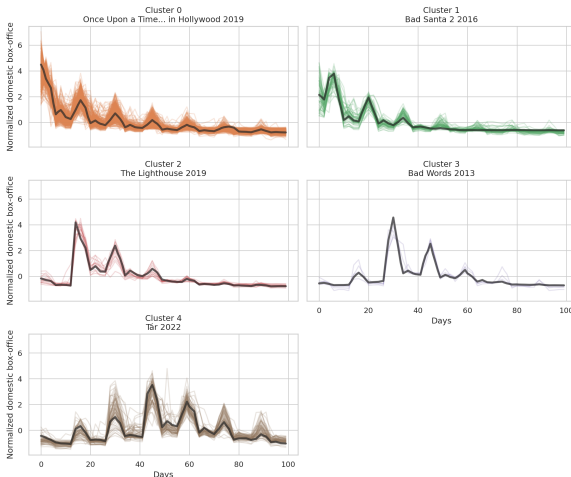


Figure 28: Clusters obtained by density-based clustering and respective cluster medoids (in black).

We then experimented with different linkage criteria for hierarchical clustering, selecting in the end average linkage, which yielded the best dendrogram (Fig. 26). After selecting an appropriate threshold, we obtained 4 clusters of very different sizes (C_1 : 75, C_2 : 658, C_3 : 101, C_4 : 15). This clustering obtained a silhouette score of 0.77.

Signals for clustered time series and cluster medoids are visualized in Fig. 27: it can be easily seen that, while C_2 and C_3 can be identified respectively with instant hits and ‘late bloomers’, the identity of movies from C_1 and C_4 is more confused. Detailed analysis of silhouette scores revealed that many records from C_4 and especially C_1 have negative silhouette score, suggesting that they were assigned to the wrong cluster.

We then run Principal Component Analysis on the dataset and visualized clusters on the two first components, that together explain more than 60% of variance (Fig. 29a, cluster medoids are showed in black).

Inspecting the PCA Eigenvectors, we found out that the first component is mainly concerned with timestamps from the first week or so, which explains why records from C_2 (and some records from C_1) have positive values in this component, while the ‘late bloomers’ from C_3 have very negative values. Also, observing the distribution of records from C_1 along the first component, it is clear why more than half of them have a negative silhouette score, since they clearly should belong to C_2 . In contrast, K-Means, clearly separates the two clusters along the first component.

We also visualized the clusters using tSNE on the DTW distance matrix. We choose this projection for its ability of preserving the local structure of the dataset. While tSNE was able to convey some information on the numeric relative dimensions of different clusters, and was able to visualize clusters nicely, in the end it proved less interesting, because of its inability of revealing the global structure of the dataset and of conveying the variable density of our clusters.

Inspection of the PCA visualization convinced us that extraction of clusters with a density-based method was viable. Since a series of run with different radii failed to return any interesting clusters, we decided to pursue a different approach: we run OPTICS (with $\text{minPts}=4$) to completion on a reduced version of the dataset, using the first 5 components created by PCA (which capture more than 80% of variance). We then visualized the reachability plot and choose an appropriate radius with which to run DBSCAN using the reachability distances computed by OPTICS. DBSCAN

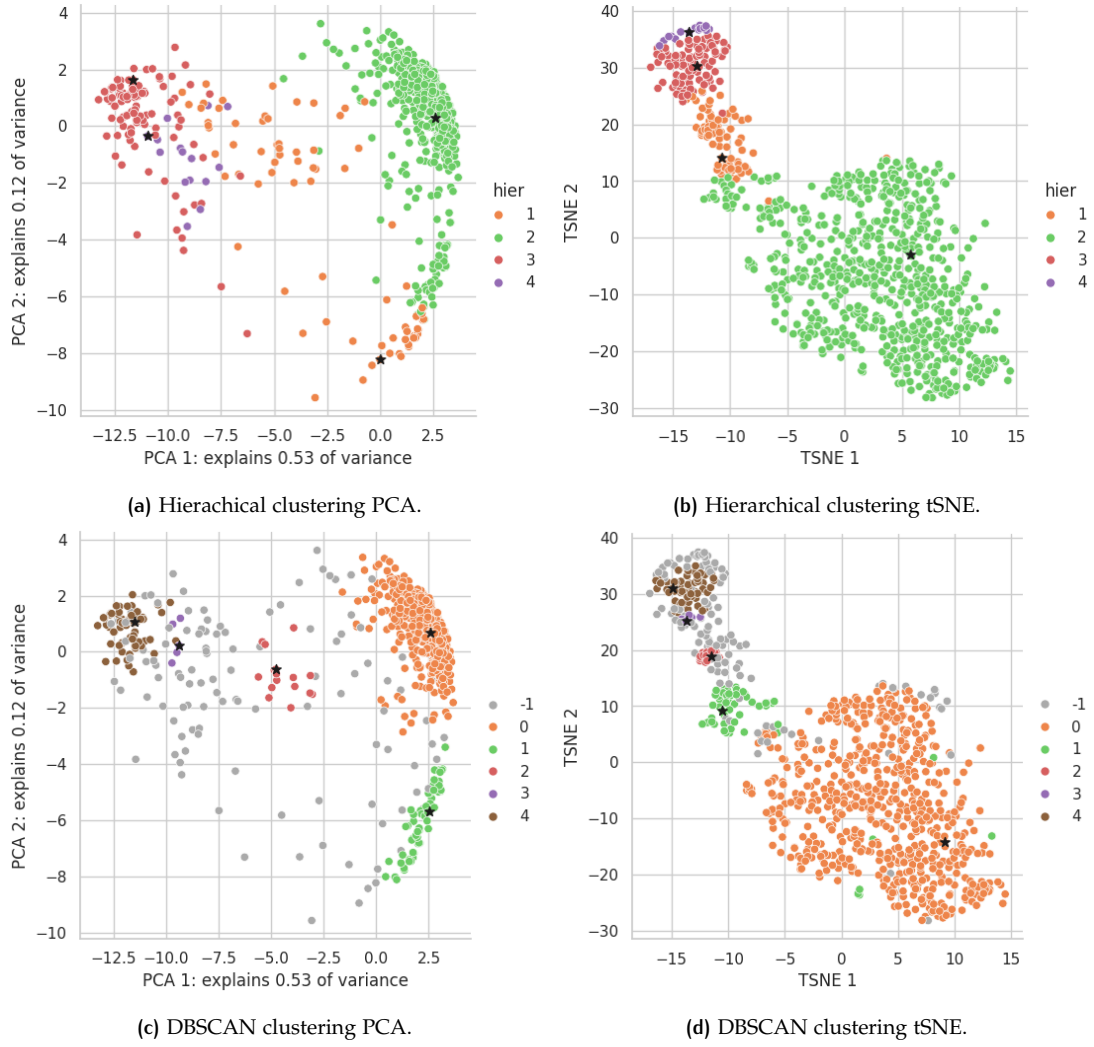


Figure 29: Clusters and cluster medoids for hierarchical clustering and DBSCAN visualized with two dimensionality reduction techniques.

detected 116 noise points and returned five clusters of varying sizes (C₀: 594, C₁:57, C₂: 16, C₃: 5 and C₄: 62, see Fig. 28).

We computed the silhouette score for non noise points on the DTW distance matrix (in order to have results comparable with the other clustering methods) and found that while globally the silhouette was a little worse than for the previous methods (0.69), essentially no record showed signs of being misclassified.

Inspection of the PCA visualization (Fig. 29c) shows that this method doesn't fall into the pitfalls of the hierarchical clustering, for example finding a good separation between C₀ and C₁.

However, while C₀ and C₁ are fully satisfactory, respectively clustering together instant hits and movies with a similar pattern to instant hits, but with a less explosive first weekend, it looks to us that the clustering of points records with lower values on the first PCA component is more con-

tentious because of the inability of DBSCAN to detecting clusters of different densities. Arguably the 'late bloomers' from C₃ and C₄ should be clustered together (probably with some noise points). C₂, despite its size, captured our interest, because it was able to detect a pattern no other clustering method was able to capture. It seems to us that movies in this cluster have a slow start at the box office, but, differently from our 'late bloomers', they explode on the first weekend after the release and then display a behavior very similar to instant hits. It is possible that movies in this cluster were pre-released in a limited fashion.

6.2.2 Distribution of shape-based clusters w.r.t. exogenous features

In this analysis we concentrated our attention on instant hits from C₃ and 'late bloomers' from C₂ returned by the hierarchical clustering. We no-

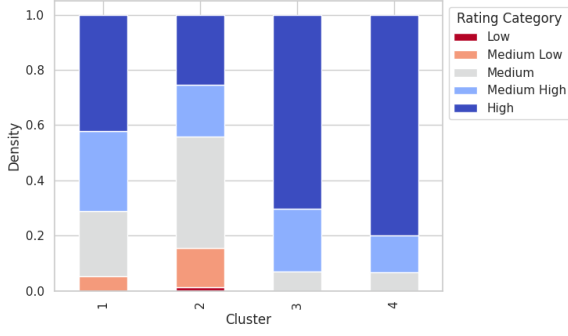


Figure 30: Distribution of binned rating by cluster (hierarchical).

ticed that ‘late bloomers’ have on average a lower average box office (median in the 100k) with respect to instant hits (median in the millions). This is a valuable finding, since we applied amplitude scaling on the time series before analysis. In addition to that, we found that ‘late bloomers’ are treated more favorably in terms of rating, as can be seen from and receive on average more awards and nominations. Action movies are severely underrepresented among ‘late bloomers’ (only 5% vs 38% of records from C2 and 32% globally), while dramatic movies are overrepresented, since, despite making up about half of the dataset, they make up more than 85% of C3.

6.2.3 Global structural features clustering

We ran hierarchical clustering on time series structural features experimenting with different linkage criteria and selecting the one that gave use the best dendrogram (Ward). We obtained four clusters and a mean silhouette of 0.21 (it should be noted that we do not invite comparison with the silhouette obtained by shape-based methods, since this clustering was performed on a wholly different distance matrix).

From Fig. 31 we can see that also this clustering method was able to detect the patterns observed previously, albeit in a less marked way. C1 and C2 (which are later merged in a single cluster) are mainly made up of ‘late bloomers’, while C3 and C4 group together instant hits (C3 and C4 are sisters).

Analysis of clusters w.r.t. exogenous features showed very similar patterns to those already observed in terms of rating, binned rating, awards, nominations, distribution of action and dramatic movies.

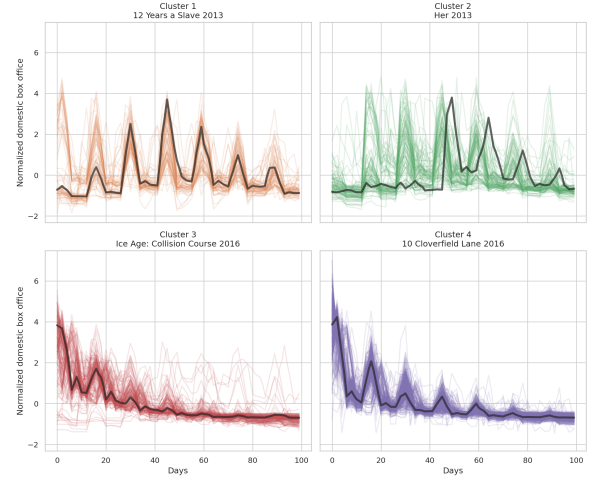


Figure 31: Clusters obtained by hierarchical clustering on structural features and respective cluster medoids (in black).

6.3 Classification and regression

For this section, we defined two classification tasks and one regression task to be approached with different representations of the time series.

The first is a binary classification task that aims to detect action movies in our dataset; for the evaluation of this task we targeted the F1 score for the positive class, as we were interested in achieving a balance between precision and recall (see table 8 for results in validation).

The second classification task we undertook is a multi-class problem consisting in predicting the binned rating of a movie; since, as we already noticed in 6.1, rating classes are imbalanced, we evaluated model performance on the basis of the macro averaged F1 score, as to guarantee acceptable performances on all classes (see table 9).

In addition to these tasks, we also experimented with some models on the regression task of predicting the mean rating of a movie (see table 10).

In order to establish that our classifiers performance was better than random, we compared results with those obtained by a dummy classifier with stratified strategy; similarly, for regression, we checked that the MSE of the models didn’t exceed the variance of the target feature (0.82). All models employed performed much better than their respective baselines.

6.3.1 Distance-based approaches

We employed KNN classifiers (and regressor) on the signal of our time series. Since no approximation was needed, the only preprocessing step

Model	precision	recall	f1	average precision
baseline	.30 ± .06	.30 ± .05	.30 ± .05	.32 ± .02
KNN	.56 ± .02	.52 ± 0.04	.54 ± .02	.58 ± .03
C22 RF	.65 ± .05	.42 ± .05	.51 ± .05	.63 ± .05
TSF	.63 ± .05	.41 ± .03	.50 ± .03	.62 ± .05
drCIF	.66 ± .04	.41 ± .08	.50 ± .06	.58 ± .05
RST	.62 ± .03	.43 ± .05	.51 ± .4	.59 ± .10
Shapelets MP df	.45 ± .04	.46 ± .08	.45 ± .06	.40 ± .04
Shapelets MP rf	.60 ± .09	.34 ± .07	.43 ± .05	.55 ± .05
Proximity Forest	.68	.40	.50	.60

Table 8: Performance of tested models for the action classification task in validation (mean and standard deviation).

Model	accuracy	precision (macro)	recall (macro)	f1 (macro)
baseline	.30 ± .02	.21 ± .01	.21 ± .01	.21 ± .01
KNN	.43 ± .03	.33 ± .06	.28 ± .02	.27 ± .02
C22 RF	.45 ± .04	.32 ± .05	.30 ± .03	.28 ± .03
TSF	.43 ± .04	.31 ± .04	.29 ± .03	.29 ± .03
drCIF	.46 ± .05	.35 ± .04	.32 ± .03	.31 ± .03
Proximity Forest	.42	.29	.27	.26

Table 9: Performance of tested models for the rating classification task in validation (macro averages).

Model	mean squared error	R2
KNN	.69 ± .11	.14 ± .07
C22 RF	.58 ± .08	.27 ± 0.9
TSF	.62 ± .01	.22 ± .08
drCIF	.60 ± .06	.23 ± .13

Table 10: Performance of tested model for the regression task.

performed on the raw signals was amplitude scaling.

For each task, we conducted grid search in order to find the optimal configuration of hyperparameters: we tested different values of k between 1 and 60 and whether to weight exemplar by distance. But the hyperparameter we were most interested in was the distance function: we considered both Euclidean Distance and two versions of constrained DTW (Sakoe-Chiba Band of radius 0.05 and 0.1). We evaluated the performance of each configuration of hyperparameters performing a 5-fold stratified cross validation on the training set. In our pursuit for the best parameters, we considered area under the PR curve for the binary classification task, macro averaged F1 for the multi-class problem and the coefficient of determination for regression. Overall our models demonstrated great sensibility to the number of neighbors, either plateauing or decreasing performance between 10 and 20 neighbors and generally

performed better when exemplars were weighted by distance.

Impact of the presence of warping and warping constraints was generally negligible, with euclidean distance performing better on the binary classification task. Even when warping impacted positively the model performance (for the multi-class problem), the improvement was negligible when considering the score variance across folds.

When evaluating our models on 5-fold stratified cross validation, we found that all of them were able to clear the respective baselines without problems. In particular, even without tuning the decision threshold our classifier obtained precision, recall and f1 all over 0.5 on the binary classification task.

The regressor demonstrated no sensibility at all to distance or weights parameters, so we picked the simplest configuration possible.

Proximity Forest is a distance based classifier that creates an ensemble of decision trees where splits are based on the similarity of the candidate time series with two exemplars measured using various distance measures. Since this model took much longer for fitting than the previous ones, we didn't tune parameters and performed hold-out validation instead of 5-fold cross validation, reserving 25% of the training test for validation.

For both classification tasks, performance was comparable with that of KNN.

6.3.2 Global structural features approaches

Our second approach was to train a classifier on global structural features extracted from the original signals. We used the same set of features we already performed clustering on (see 6.2.3) and used Random Forest Classifier/Regressor as an estimator, a model we chose for its ability to provide somewhat interpretable results (by inspection of feature importance) while being less prone to overfitting than Decision Trees.

When tuning hyperparameters, we let individual trees fully develop and manipulated only the number of estimators and the maximum number of features used by each tree: we experimented with 10, 50, 100 and 200 estimators and with the following values of features: `log2`, `sqrt`, `all features`. Grid search was performed with the same modalities and target scores detailed in 6.3.1.

While 10 estimators proved too few, overall the models didn't show an extreme sensitivity to hyperparameters: standard deviation across folds was generally greater than differences between parameters configurations. 50 estimators and all features were found to be the best configuration for both multi-class classification and regression, while for binary classification we used 200 estimators and `max_features='log2'`.

On cross-validation these models outperformed KNN on all tasks, albeit non decisively, except on the regression task. While F1 score on the binary classification task was lower than that of KNN, area under the PR curve was greater, indicating that KNN could be out-performed with decision threshold tuning.

6.3.3 Ensemble of interval-based approaches

These approaches operate on representations of the original time series that are created by extracting summary statistics from selected subsequences. We experimented with Time Series Forest, a very simple model that summarizes intervals only by mean, standard deviation and slope, and with Diverse Representation CIF (DrCIF), which represents an extension of Canonical Interval Forest and is generally considered the state of the art for this family of approaches.

For both TSF and drCIF, we experimented with different combinations of minimum and maximum interval length. For the minimum interval length we capitalized on insight gained from motifs discovery, so we tested with lengths of 6, 7, 8, 14, 28 and 35 timestamps. Generally speaking, a smaller interval length (7) was preferred for all tasks. As for the maximum interval length,

we allowed for either half the size of the time series, or the whole time-series. We noticed that the influence of this second parameter became non-existent the more estimators we considered: for TSF we experimented with 5, 10, 50, 100 and 200 estimators, while for drCIF, which is much more computationally expensive, we limited the search to 5 and 10 estimators. While the impact of number of estimators is not clear on the binary classification task, we noticed improvements on both the multi-class problem and the regression when using more estimators.

On 5-fold cross validation, TSF and drCIF demonstrated performances comparable to those of the previous approach. The biggest gap is on the regression task, but considering the variance across scores from different folds, this difference is probably not significant. It is possible that drCIF performance, which is practically indistinguishable from that of TSF, was hindered by our choice to limit the number of estimators considered.

6.3.4 Shapelets

6.3.5 Assessment